SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I **Sasi Kumar. S(192324254)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Sasi Kumar. S**

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

DETAILED TECHNICAL SOLUTIONS & ARCHITECTURAL DESIGNS

Scenario 1: Media Company Video Streaming Platform Cloud Migration

Problem Context: A media enterprise seeks to migrate its high-throughput video streaming infrastructure to the public cloud. The target architecture must seamlessly ingest, transcode, and stream high-definition content to millions of concurrent viewers with sub-second latency, zero buffering stalls, and optimized operational expenditure.

1. Cloud-Native Scalability & Multi-Tier Caching Architecture

To support millions of concurrent streaming sessions without playback interruptions, the media platform decouples compute ingestion from delivery. The system leverages Adaptive Bitrate Streaming (ABR) via HTTP Live Streaming (HLS) and Dynamic Adaptive Streaming over HTTP (DASH):

- **Ingestion & Distributed Transcoding Pipeline:** High-bitrate mezzanine video source files are uploaded via Amazon S3 Transfer Acceleration to an ingestion bucket. Upload events trigger an event-driven workflow using AWS Step Functions and AWS Elemental MediaConvert (or containerized FFmpeg spot worker nodes on AWS EKS). The pipeline segments video into discrete 2- to 6-second fragments (.ts / fragmented MP4) across multiple quality ladders (1080p, 720p, 480p, 360p) and compiles adaptive manifest files (.m3u8 / .mpd).
- **Hierarchical Three-Tier Caching Architecture:** To prevent origin storage bottlenecks, content delivery employs three distinct caching boundaries: (1) Global Edge PoP Caching via

Amazon CloudFront CDN, positioning cached video fragments directly within ISP networks to deliver >96% cache-hit ratios; (2) CloudFront Origin Shield, a centralized regional caching tier placed between edge locations and origin S3 storage to aggregate cache misses and eliminate the stampeding herd effect; and (3) In-Memory Dynamic Caching (Amazon ElastiCache Redis Cluster) deployed in the application tier to store user watch progress, authorization tokens, and video metadata with sub-millisecond response times.

2. Load Balancing and Dynamic Auto-Scaling for Smooth Streaming

- **Layer-7 Application Load Balancing (ALB):** Incoming user requests are routed through dual-stack (IPv4/IPv6) ALBs deployed across three Availability Zones (AZs). ALBs execute SSL/TLS offloading using TLS 1.3 and route traffic using path-based rules: authentication and payment transactions are steered to secure compute pools, while catalog search and player manifest queries route to high-throughput compute clusters. Real-time synthetic health checks isolate unhealthy targets within 5 seconds.
- **Predictive and Target Tracking Auto-Scaling:** Stateless playback API services running on Amazon Elastic Kubernetes Service (EKS) scale horizontally using the Horizontal Pod Autoscaler (HPA) paired with Kubernetes Cluster Autoscaler. During scheduled live events (e.g., sporting championships), AWS Predictive Scaling evaluates historical viewership cycles and pre-allocates compute capacity 25 minutes prior to peak demand, bypassing VM cold-boot latency.
- **Connection Draining & Graceful Session Handoff:** During scale-in events, ALBs enforce deregistration delays (connection draining), allowing active TCP streaming connections to complete gracefully without client disconnects or buffering errors.

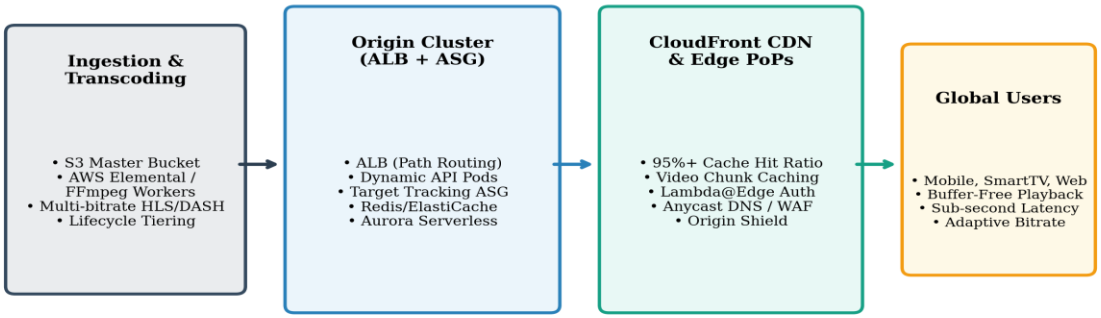


Figure 1: Cloud-Native Video Streaming Architecture: Ingestion, Multi-Bitrate Transcoding, Origin Shield, and Global CDN Delivery.

3. Strategic Cost Optimization for High-Demand Streaming

- **Data Egress Elimination via Edge Caching:** Direct public downloads from cloud object storage incur standard internet egress fees ($0.09/GB). By serving >96% of video segments from CloudFront edge caches, data transfer between S3 and CloudFront is free ($0.00/GB), and negotiated CDN committed-use tiers reduce overall bandwidth delivery costs by over 60%.
- **Automated Storage Lifecycle Management:** S3 Intelligent-Tiering automatically transitions newly released titles from S3 Standard to S3 Infrequent Access (IA) after 30 days of declining viewership, and deep catalog archives transition to S3 Glacier Flexible/Deep Archive ($0.00099/GB/month), yielding up to 70% storage expenditure reduction.
- **EC2 Spot Instances for Transcoding Queues:** Because media transcoding jobs are stateless, chunk-based batch operations, running transcoding worker pods on AWS Spot Instances delivers up to 90% cost savings compared to On-Demand compute instances.

Scenario 2: Multi-Cloud Container Orchestration for Fintech Startup

Problem Context: A growing fintech enterprise deploys containerized financial microservices across multiple cloud service providers (AWS and Azure). The engineering team faces critical challenges in managing configuration drift, ensuring unified service discovery, maintaining high availability, and adhering to banking regulatory mandates.

1. Container Orchestration & Unified Multi-Cloud Solution

- **Standardized Kubernetes Abstraction:** To prevent cloud provider lock-in, all applications are packaged strictly as Open Container Initiative (OCI) images and orchestrated using managed Kubernetes services: AWS Elastic Kubernetes Service (EKS) for core transaction pipelines and Azure Kubernetes Service (AKS) for ledger analytics and disaster failover.
- **Declarative GitOps Deployment Framework:** To eliminate environment inconsistencies and manual configuration drift, the startup adopts a GitOps deployment methodology using ArgoCD. A centralized Git repository maintains version-controlled Helm charts and Kustomize overlays. When code merges to the production branch, ArgoCD reconciliation agents inside each cluster automatically pull and apply identical deployment states, security contexts, and resource quotas.
- **Infrastructure as Code (IaC):** Terraform automates multi-cloud provisioning, ensuring network CIDR blocks, VPC peering links, and cloud-native IAM security boundaries remain deterministic.

2. Cross-Cloud Service Discovery, Mesh Communication, and Elastic Scaling

- **Global Multi-Cluster Service Mesh (Istio / Consul):** Standard internal DNS cannot span disparate cloud provider VPCs. The architecture deploys an Istio Multicluster Service Mesh connecting AWS EKS and Azure AKS over private IPsec VPN / DirectConnect

interconnects. Istio Ingress and Egress gateways enforce Mutual TLS (mTLS) with SPIFFE/SPIRE cryptographic identities, ensuring end-to-end data encryption in transit. Microservices discover each other using uniform virtual domain naming (e.g., 'payment.service.fintech.mesh').

- **Two-Tier Horizontal & Vertical Scaling:** At the application layer, Kubernetes Horizontal Pod Autoscaler (HPA) scales microservice pods dynamically based on incoming transaction volume and queue latency. At the compute layer, Karpenter / Cluster Autoscaler provisions bare-metal and VM worker nodes across multi-cloud pools to eliminate compute starvation.
- **Global Anycast Traffic Management:** Cloudflare Magic Transit and Anycast Global Server Load Balancing (GSLB) route user transaction traffic to the geographically nearest healthy cloud cluster while monitoring real-time API latency and regional uptime.

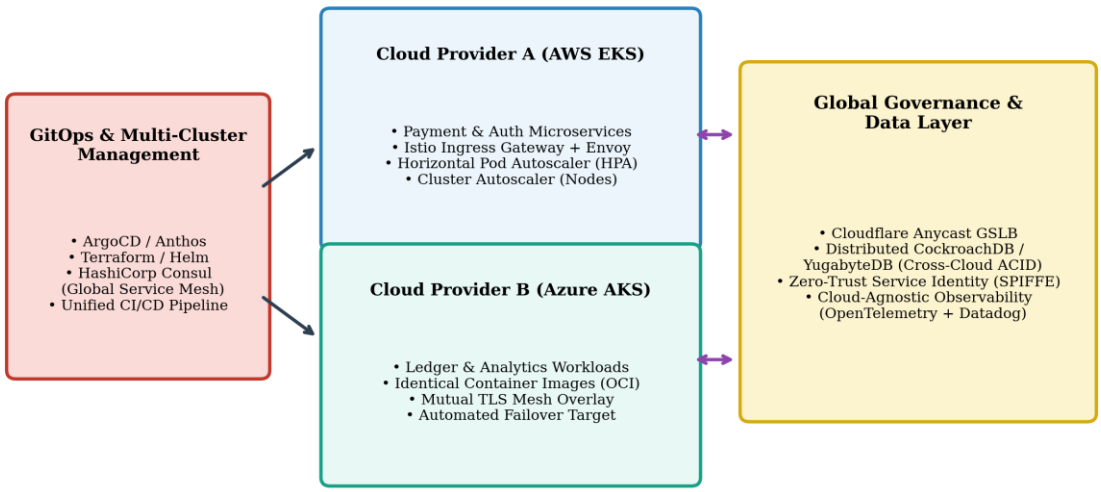


Figure 2: Multi-Cloud Fintech Container Orchestration: GitOps Declarative Ingress, Istio Service Mesh, and Global Anycast Routing.

3. Comprehensive Multi-Cloud Evaluation: Strategic Benefits vs. Architectural Risks

Strategic Benefits	Architectural Risks & Operational Challenges
Eradication of Vendor Lock-In: Eliminates single-vendor commercial dependency, providing pricing negotiation leverage and workload mobility.	Cross-Cloud Data Egress Overhead: Replicating transactional state and ledger data across providers incurs high cross-WAN bandwidth egress fees ($0.09/GB).
Regulatory Resilience: Meets strict banking authority mandates (e.g., RBI, DORA) requiring secondary non-correlated infrastructure for disaster survival.	Expanded Threat Surface: Managing disparate IAM permissions, security group semantics, and compliance audit frameworks increases operational complexity.
Best-of-Breed Feature Selection: Enables pairing AWS's mature transaction relational database engines	Operational Complexity & Skill Scarcity: Requires site reliability engineers to maintain deep architectural

with Azure's enterprise Active Directory ecosystem.	proficiency across multiple hyperscaler platforms.
Total Cloud Outage Immunity: Major availability zone or regional service failures in AWS US-East-1 do not affect transaction processing on Azure.	Distributed State Latency: Achieving ACID database consistency across multi-cloud clusters introduces physical network round-trip time latency.

Scenario 3: E-Learning Platform Disaster Recovery and Data Protection

Problem Context: An e-learning provider suffered severe data loss caused by accidental deletion of educational content, student submission records, and database tables. The platform requires an immutable, cloud-native backup and disaster recovery architecture capable of satisfying stringent Recovery Point Objectives (RPO) and Recovery Time Objectives (RTO).

1. Cloud-Native Backup, Versioning, and Data Protection Solution

- **S3 Object Versioning & Delete Marker Recovery:** All educational asset buckets enforce S3 Object Versioning. In a version-enabled bucket, accidental or unauthorized HTTP 'DELETE' requests do not erase underlying data; they merely append a soft 'Delete Marker'. Prior object versions remain cryptographically preserved and can be restored instantaneously via automated Boto3 Python scripts.
- **MFA Delete & WORM Object Lock (Compliance Mode):** To prevent rogue scripts or compromised administrative credentials from permanently purging version histories, MFA Delete is mandated via hardware security tokens. Furthermore, finalized academic records, grades, and certifications are placed in S3 Object Lock under Compliance Mode. Once locked, files cannot be overwritten or deleted by any user—including AWS root account credentials—until the predefined retention timer expires.
- **Continuous Database Point-in-Time Recovery (PITR):** Relational course management databases (Amazon Aurora PostgreSQL) maintain continuous automated write-ahead redo log backups alongside daily system snapshots. If an engineer executes an accidental 'DROP TABLE' query at 11:15:32 AM, the cluster can be restored to 11:15:31 AM with zero data corruption.
- **Cross-Region Air-Gapped Replication (CRR):** S3 Cross-Region Replication automatically mirrors all backup payloads asynchronously to a secondary cloud region within a separate, air-gapped AWS account governed by isolated IAM access perimeters.

2. Achieving Stringent RTO and RPO Targets

- **Recovery Point Objective (RPO <= 5 Minutes):** RPO represents the maximum tolerable data loss duration. The architecture fulfills an RPO under 5 minutes through continuous Aurora transaction log streaming (replication lag < 1 second) and S3 Replication Time Control (S3 RTC), which SLA-guarantees byte-level object replication to the secondary region within 15 minutes, with typical real-world replication completing in under 45 seconds.

- **Recovery Time Objective (RTO <= 30 Minutes):** RTO represents the maximum acceptable system downtime. The platform deploys a Pilot Light architecture in the secondary recovery region. Core VPC infrastructure, load balancers, and a dormant Aurora read replica remain pre-provisioned. Upon disaster declaration, automated Terraform pipelines promote the read replica to standalone primary status (under 2 minutes) and hydrate EKS container worker nodes (under 12 minutes), restoring full operational capability in less than 15 minutes.

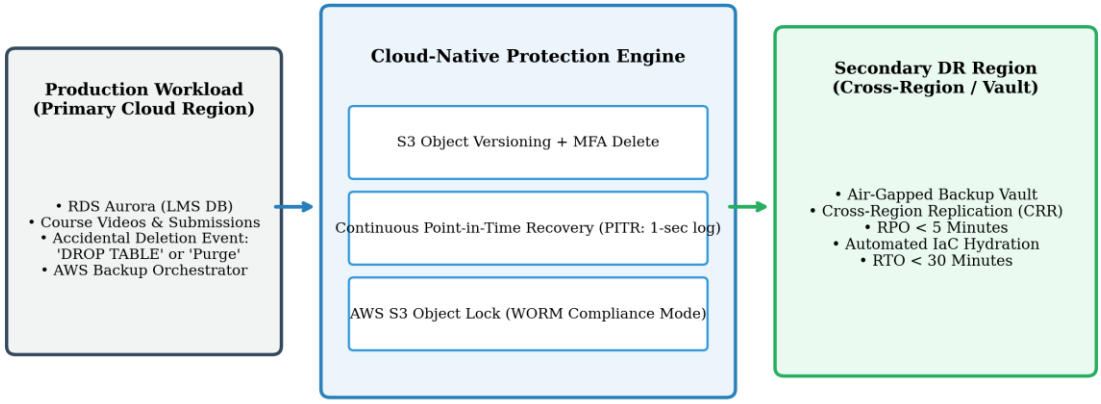


Figure 3: Comprehensive Cloud Disaster Recovery Pipeline: S3 Object Versioning, Continuous Database PITR, and Cross-Region Vault Replication.

3. Step-by-Step Business Continuity Execution Framework

- **Step 1 - Anomaly Detection:** AWS CloudWatch Alarms and EventBridge detect sudden 4xx/5xx HTTP error surges or database drop events, paging on-call site reliability engineers via PagerDuty.
- **Step 2 - Blast Radius Containment:** The primary API Gateway is placed into read-only maintenance mode to prevent corrupt transactions from compounding database inconsistencies.
- **Step 3 - Automated Storage & Database Recovery:** For object storage, automated lambda functions scan for recent Delete Markers and remove them, instantly unhiding deleted assets. For database drops, Aurora executes point-in-time restoration to the exact second prior to the deletion timestamp.
- **Step 4 - Traffic Rerouting & DNS Failover:** Amazon Route 53 health checks update public DNS records, shifting global student traffic to the hydrated recovery cluster.
- **Step 5 - Forensic Audit & Preventive Hardening:** AWS CloudTrail audit logs are inspected to identify the source of the deletion command. IAM least-privilege boundaries and

SCP delete-prevention policies are reinforced to prevent recurring human or programmatic errors.

Scenario 4: Government Agency Hybrid Cloud Architecture and Governance

Problem Context: A national government agency must modernize public digital citizen services while maintaining strict compliance with defense and privacy regulations. Classified intelligence databases and citizen master records must remain housed inside on-premises datacenters, while citizen-facing web portals and tax filing services are deployed to the elastic commercial public cloud.

1. Secure Hybrid Network Connectivity Architecture

- **Dedicated High-Bandwidth Private Circuit:** Communication between the government on-premises datacenter and cloud VPCs is established via redundant 10 Gbps AWS Direct Connect (or Azure ExpressRoute) circuits. This private fiber interconnect bypasses public internet transit entirely, guaranteeing predictable latency and high throughput.
- **Defense-in-Depth Hardware Encryption (MACsec + IPsec):** To safeguard against physical cable wiretapping, IEEE 802.1AE MACsec (Media Access Control Security) delivers 256-bit point-to-point line-rate encryption at Layer 2. Overlaid on this is an IPsec VPN tunnel terminating inside an AWS Transit Gateway (TGW) with BGP dynamic routing, ensuring multi-layer cryptographic isolation.
- **DMZ Perimeter Inspection & Egress Filtering:** All cross-boundary queries traversing between cloud and on-premises environments are routed through a transit DMZ containing Next-Generation Web Application Firewalls (WAF) with Deep Packet Inspection (DPI) and Intrusion Prevention Systems (IPS) to prevent lateral unauthorized access.

2. Unified Identity Management, Access Control, and Key Governance

- **Federated Single Sign-On (SSO) with Multi-Factor Authentication:** The agency maintains on-premises Microsoft Active Directory as the authoritative Identity Provider (IdP). Cloud access is integrated via SAML 2.0 / OIDC federation using AWS IAM Identity Center. Government personnel authenticate using hardware FIDO2 Smart Cards (CAC/PIV) and biometric MFA, receiving short-lived, role-based Security Token Service (STS) credentials that eliminate hardcoded API keys.
- **Dedicated Hardware Security Modules (CloudHSM):** All data stored in the cloud is encrypted at rest using AES-256 via Customer Managed Keys (CMKs). Keys are generated, stored, and managed exclusively inside dedicated FIPS 140-2 Level 3 Hardware Security Modules (AWS CloudHSM). The cryptographic keys never leave physical tamper-resistant silicon boundaries, ensuring cloud hypervisor personnel cannot access plaintext citizen records.

- **Mandatory In-Transit Encryption:** All microservice-to-database communications enforce TLS 1.3 with Mutual Authentication (mTLS) using certificates issued by a private Certificate Authority (AWS Private CA).

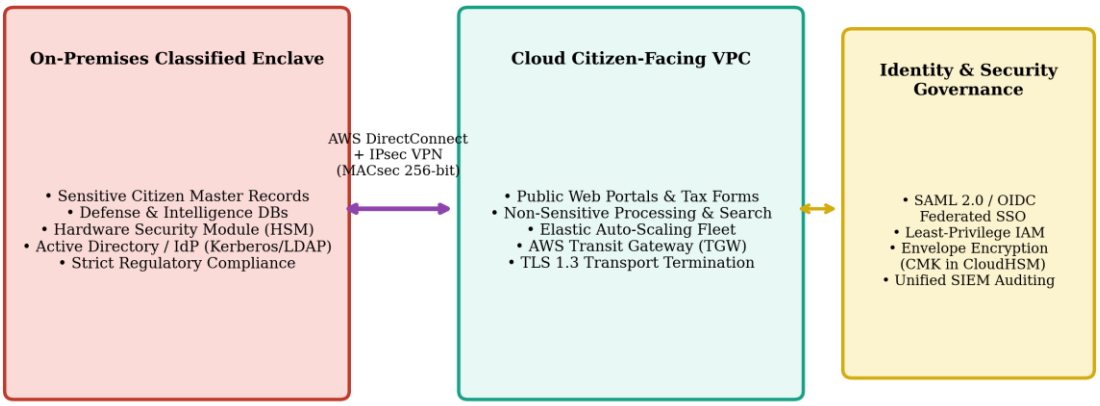


Figure 4: Secure Government Hybrid Cloud Architecture: MACsec/IPsec Interconnect, Centralized IAM Federation, and CloudHSM Cryptography.

3. Key Challenges and Mitigation in Hybrid Cloud Management

- **Network Latency Asymmetry & The 'Chatty API' Bottleneck:** When cloud web applications perform numerous synchronous database queries to on-premises databases across a hybrid pipe, network latency accumulates rapidly. The agency mitigates this by refactoring applications to use asynchronous message queues (Amazon SQS), edge caching, and coarse-grained GraphQL/REST endpoints.
- **Geopolitical Data Sovereignty:** Strict Service Control Policies (SCPs) restrict cloud provisioning exclusively to domestic geographic regions (e.g., AWS ap-south-1 Mumbai), blocking any API calls attempting to transmit citizen records beyond national physical borders.
- **Fragmented Observability & Siloed Logging:** Managing disparate logging tools across environments creates compliance blind spots. The solution deploys an enterprise SIEM (Splunk Enterprise) that aggregates both cloud VPC Flow Logs and on-premises intrusion logs into an immutable, unified audit dashboard.
- **Split-Brain DNS Resolution:** Amazon Route 53 Resolver Endpoints bridge cloud private hosted zones with internal on-premises Active Directory DNS, ensuring bidirectional name resolution.

Scenario 5: Multi-Region High Availability and Fault Resiliency for SaaS Provider

Problem Context: A global enterprise SaaS vendor requires 99.999% availability ('five-nines') and total disaster resilience. The application must seamlessly withstand catastrophic cloud region outages (such as major optical cable severing or extreme weather events) without customer service disruption or permanent data loss.

1. Multi-Region Active-Active / Warm-Standby Deployment Strategy

- **Stateless Compute Layer (Active-Active Multi-Region):** Containerized microservice pods running on Amazon EKS are deployed concurrently across two geographically separated cloud regions (Primary: US-East-1 Virginia, Secondary: US-West-2 Oregon). Microservice fleets in both regions actively process user requests, minimizing physical network latency for global clients.
- **Stateless Session Management:** User sessions are decoupled from individual instances using stateless JSON Web Tokens (JWT) and synchronized across regions via Amazon ElastiCache Global Datastore for Redis, allowing users to transition between cloud regions without session drops or forced re-login.
- **Global Traffic Steering:** AWS Global Accelerator and Route 53 Traffic Flow announce Anycast static IP addresses from edge locations worldwide, dynamically routing user requests to the closest healthy region based on real-time network path telemetry and latency probes.

2. Multi-Region Data Replication and Automated Failover Mechanics

- **Storage-Level Asynchronous Replication (Amazon Aurora Global Database):** Primary relational transactions execute in Region 1. Dedicated storage engines stream write redo logs directly across AWS's private optical network to read-only replica clusters in Region 2 with typical replication latency under 1 second. Because replication occurs at the storage layer, primary database compute performance remains unimpacted.
- **Cross-Region Object Replication (S3 CRR):** Customer file uploads and multi-tenant assets are mirrored using S3 Cross-Region Replication with Replication Time Control (RTC), guaranteeing byte-level consistency across regions within 15 minutes and SLA backing.
- **Automated Failover Orchestration via Route 53 ARC:** Amazon Route 53 Application Recovery Controller (ARC) continuously evaluates synthetic end-to-end health canaries. If Region 1 experiences a total failure, ARC triggers an automated regional failover: the secondary Aurora replica is promoted to primary read/write status in under 1 minute (achieving RPO < 1 sec, RTO < 1 min), while Route 53 shifts 100% of global DNS traffic to Region 2.

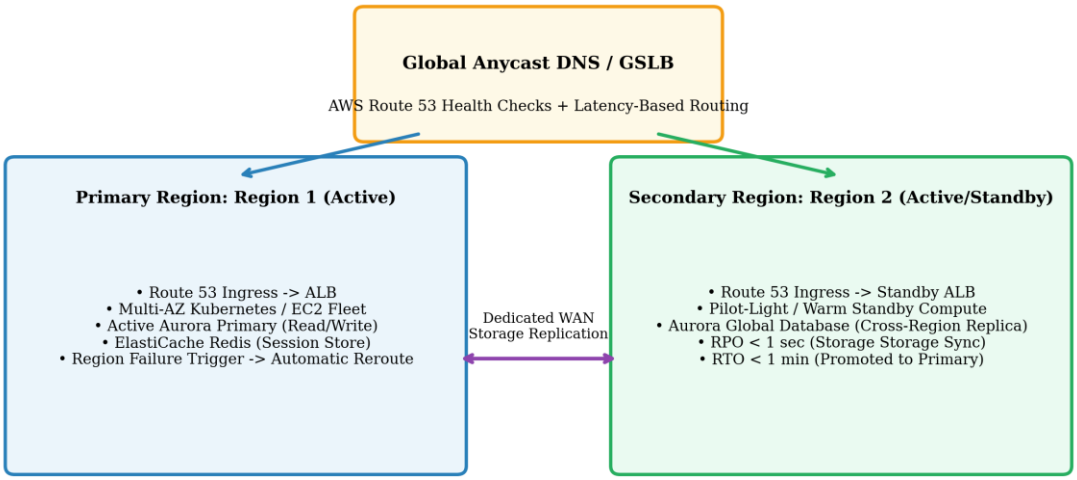


Figure 5: Multi-Region SaaS Fault-Tolerant Architecture: Anycast GSLB Ingress, Cross-Region Aurora Storage Sync, and Automated Failover.

3. Proactive Telemetry, Synthetic Canaries, and Self-Healing Alerting

- **Full-Stack Distributed Telemetry (OpenTelemetry):** The platform collects metrics, structured logs, and distributed traces using OpenTelemetry standards. AWS CloudWatch monitors infrastructure saturation, Fluentbit ships JSON logs to OpenSearch, and AWS X-Ray traces distributed request lifecycles to detect performance bottlenecks prior to failure.
- **Synthetic Canary Probing:** CloudWatch Synthetics executes 24/7 automated headless browser canaries from global vantage points, simulating end-user workflows (user login, subscription checkout, dashboard generation). Any failure triggers proactive alerts before impacting real customers.
- **Automated Self-Healing Runbooks:** CloudWatch Alarms integrate with AWS Systems Manager Automation and Lambda runbooks to automatically terminate unresponsive pods, recycle deadlocked workers, or scale read replicas without human intervention.